Amino acids and copper both serve vital roles in many biological processes of living organisms. Beyond their separate biological roles, however, chemists have also discovered that amino acids are able to function as chelates with Cu^{2+} ions to form blue complexes with potential medicinal applications. One such complex, copper(II) glycinate, has shown to be effective against some oral diseases. The reaction to form copper(II) glycinate is shown in Figure 1.

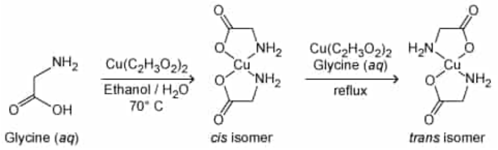


Figure 1 Synthesis of two isomers of copper(II) glycinate

Two different isomers can form depending on the reaction conditions. Mixing hot aqueous glycine (70 °C) with a hot solution of copper(II) acetate dissolved in 50% aqueous ethanol, followed by cooling the mixture on ice, yields a precipitate of only the *cis* isomer. If a suspension of the *cis* isomer is held at boiling in an aqueous solution of glycine and copper(II) acetate, the *trans* isomer is obtained.

Because of the geometric differences between the two isomers, the stretching vibrations of the central Cu–N and Cu–O bonds in each isomer are different. Symmetric and asymmetric bond stretching give four possible vibrational modes for each isomer, as shown in Figure 2.

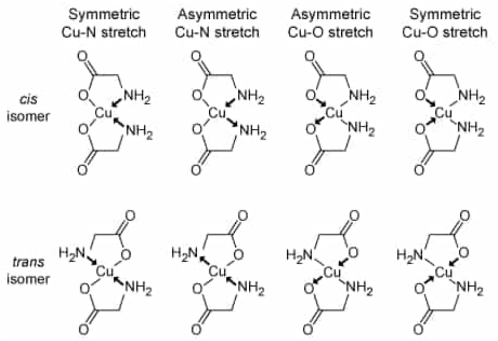


Figure 2 Vibrational modes of the coordinate bonds in copper(II) glycinate

If the stretching of the bonds during vibration causes a transient, unequal distribution of electron density between the atoms within the bond, an oscillating dipole will form. Bonds with dipoles that are not canceled out by another dipole in the molecule will absorb infrared light at particular frequencies (often expressed in wave numbers) and are designated as IR-active. As a result, the vibrational differences between the two isomers can be analyzed by far-infrared (far-IR) spectroscopy, as shown in Figure 3.

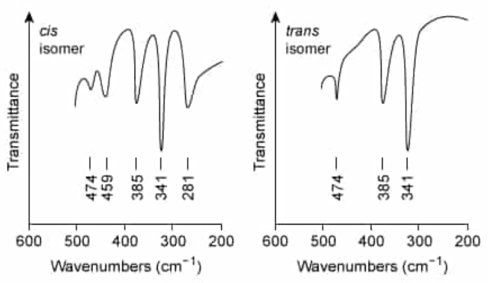


Figure 3 Far-IR spectra of the *cis* and *trans* isomers of copper(II) glycinate

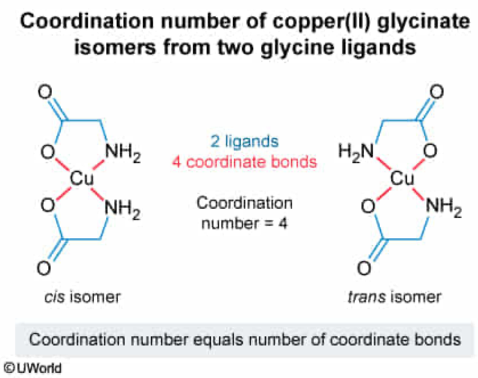
Both isomers show a skeletal absorbance (unrelated to coordinate bond vibration) near 385 cm^{-1}. Additional absorbance bands are seen that specifically relate to the vibrational modes of the bonds in each isomer.

In metal complexes such as copper(II) glycinate, the coordination number refers to:

- A. the number of ligand molecules that form the complex. [15%]
- ✔ B. the number of coordinate bonds formed. [48%]
- C. the number of electrons involved in coordinate bonding. [8%]
- D. the charge of the metal ion. [27%]

Correct
48% answered correctly

Explanation:



Coordinate covalent bonds, unlike covalent bonds, are formed between two atoms when *both* shared electrons are donated by the same atom. Such coordinate bonds are often formed between electron-poor metal ions and molecules called **ligands** that contain one or more electron-rich atoms with available lone-pair electrons. The coordinately bonded metal and its ligands are called a **complex**.

As originally defined by Alfred Werner, the coordination number of a metal complex refers to the **number of coordinate bonds** formed between the central metal ion and its nearest neighboring atoms. When all these nearest neighboring atoms are from *separate* molecules or ions, the number of ligands will equal the coordination number. However, if two or more of these nearest neighboring atoms are joined to the *same* coordinating ligand unit, then the number ligands will not equal the coordination number. In both cases, the number of nearest neighboring atoms and coordinate bonds is unchanged, but the number of ligand units is different.

For example, if bonding interactions from other ligands such as water are excluded (as shown in Figure 1), both isomers of the copper(II) glycinate complex have a coordination number of 4. The Cu^{2+} atom has two oxygen atoms and two nitrogen atoms as its nearest neighbors, forming four coordinate bonds. However, each pair of N and O atoms is joined to the same glycine molecule. As such, the four neighboring atoms come from only two glycine ligands.

(Choice A) Copper(II) glycinate has only two ligands of glycine, but its coordination number is 4.

(Choice C) The total number of electrons involved in the bonding does not directly describe the resulting coordination to the metal center.

(Choice D) The charge on a metal atom is determined by its oxidation state and bonding configuration, but the magnitude of the charge does not necessarily match the coordination number of the ligands to the metal center. Note that the "(II)" in the name of the compound refers to the oxidation state of the copper atom and does not refer to the number of ligands or to the coordination number.

Educational objective:
Coordinate covalent bonds are formed between two atoms when one atom donates both shared electrons. Such bonds are often formed between electron-poor metal ions and electron-rich atoms in ligands in a complex, and the coordination number of the complex refers to the number of coordinate bonds formed with the metal ion.

Time spent:0 sec
Subject:
General Chemistry

QID:402278
System:
Interactions of Chemical Substances

Amino acids and copper both serve vital roles in many biological processes of living organisms. Beyond their separate biological roles, however, chemists have also discovered that amino acids are able to function as chelates with Cu^{2+} ions to form blue complexes with potential medicinal applications. One such complex, copper(II) glycinate, has shown to be effective against some oral diseases. The reaction to form copper(II) glycinate is shown in Figure 1.

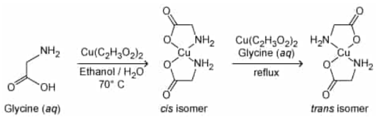


Figure 1 Synthesis of two isomers of copper(II) glycinate

Two different isomers can form depending on the reaction conditions. Mixing hot aqueous glycine (70 °C) with a hot solution of copper(II) acetate dissolved in 50% aqueous ethanol, followed by cooling the mixture on ice, yields a precipitate of only the *cis* isomer. If a suspension of the *cis* isomer is held at boiling in an aqueous solution of glycine and copper(II) acetate, the *trans* isomer is obtained.

Because of the geometric differences between the two isomers, the stretching vibrations of the central Cu–N and Cu–O bonds in each isomer are different. Symmetric and asymmetric bond stretching give four possible vibrational modes for each isomer, as shown in Figure 2.

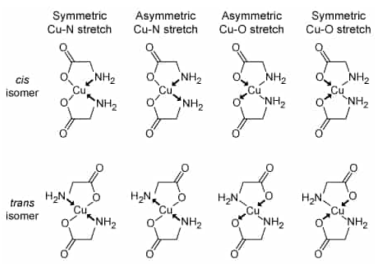


Figure 2 Vibrational modes of the coordinate bonds in copper(II) glycinate

If the stretching of the bonds during vibration causes a transient, unequal distribution of electron density between the atoms within the bond, an oscillating dipole will form. Bonds with dipoles that are not canceled out by another dipole in the molecule will absorb infrared light at particular frequencies (often expressed in wave numbers) and are designated as IR-active. As a result, the vibrational differences between the two isomers can be analyzed by far-infrared (far-IR) spectroscopy, as shown in Figure 3.

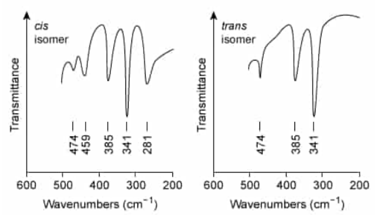
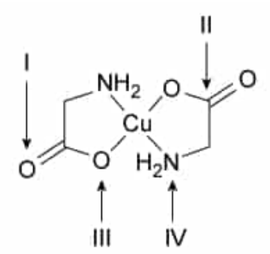


Figure 3 Far-IR spectra of the *cis* and *trans* isomers of copper(II) glycinate

Both isomers show a skeletal absorbance (unrelated to coordinate bond vibration) near 385 cm^{-1}. Additional absorbance bands are seen that specifically relate to the vibrational modes of the bonds in each isomer.

The copper(II) glycinate isomers contain atoms with different hybridization states.

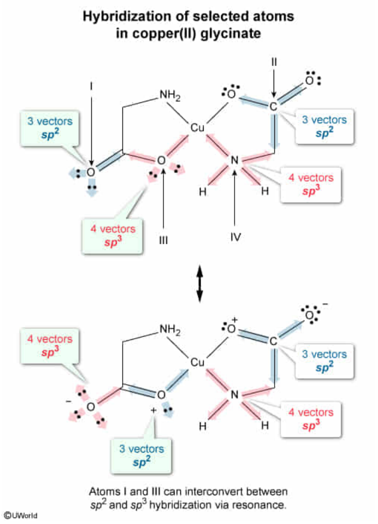


In the structure of the *trans* isomer, which of the labeled atoms has both a partial sp^2 and sp^3 hybridized character due to resonance?

- A. I and II only [26%]
- B. I and III only [49%]
- C. III and IV only [10%]
- D. II and IV only [13%]

Correct
49% answered correctly

Explanation:



Hybridization is the process by which two or more atomic orbitals combine to form new **hybrid orbitals** that have different shapes and spatial orientations than the atomic orbitals that produce them. Different types of hybrid orbitals are produced depending on the **number** and **type** of **orbitals** that **combine**.

To **minimize electronic repulsion**, hybrid orbitals adopt specific orientations that are determined by the number of hybrid orbitals formed. As such, the type of hybridization can be determined by the number of unique orbital **vectors** (directions) taken by bonds and/or lone pair electrons around each atom.

However, compounds with **resonance** (ie, delocalized electrons) cannot be fully described using a single Lewis structure. In resonance, the arrangement of the atoms is unchanged, but some π bonds and nonbonding electrons are "spread out" between multiple atoms. Consequently, atoms participating in resonance may exhibit a blend of different hybridization states in the resonance hybrid.

In the given structure of *trans*-copper(II) glycinate, atom I has 3 electron domain vectors (ie, 1 C=O bond and 2 nonbonding electron pairs) and is sp^2 hybridized. In contrast, atom III has 4 electron domain vectors (ie, 2 C–O bonds and 2 nonbonding electron pairs) and is sp^3 hybridized. However, both atoms I and III can participate in resonance to reconfigure their number of π bonds and nonbonding electron pairs. Therefore, atoms I and III have both a partial sp^2 and sp^3 hybridized character due to resonance.

(Choice A) Although atom II can participate in resonance, its hybridization is sp^2 (ie, 3 electron domain vectors) and remains unchanged in the process.

(Choices C and D) Atom IV cannot participate in resonance and remains sp^3 hybridized (ie, 4 electron domain vectors).

Educational objective:
During hybridization, atomic orbitals combine to form new hybrid orbitals that assume specific orientations to minimize the electronic repulsion between orbitals. The number of hybrid orbitals that form and the number of orientational vectors adopted by the hybrid orbitals will equal the number of atomic orbitals that combine. However, hybridization can be altered via resonance with neighboring atoms.

Time spent: 0 sec
Subject: General Chemistry

QID: 402279
System: Interactions of Chemical Substances

Amino acids and copper both serve vital roles in many biological processes of living organisms. Beyond their separate biological roles, however, chemists have also discovered that amino acids are able to function as chelates with Cu^{2+} ions to form blue complexes with potential medicinal applications. One such complex, copper(II) glycinate, has shown to be effective against some oral diseases. The reaction to form copper(II) glycinate is shown in Figure 1.

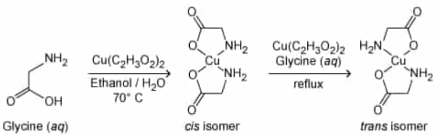


Figure 1 Synthesis of two isomers of copper(II) glycinate

Two different isomers can form depending on the reaction conditions. Mixing hot aqueous glycine (70 °C) with a hot solution of copper(II) acetate dissolved in 50% aqueous ethanol, followed by cooling the mixture on ice, yields a precipitate of only the *cis* isomer. If a suspension of the *cis* isomer is held at boiling in an aqueous solution of glycine and copper(II) acetate, the *trans* isomer is obtained.

Because of the geometric differences between the two isomers, the stretching vibrations of the central Cu–N and Cu–O bonds in each isomer are different. Symmetric and asymmetric bond stretching give four possible vibrational modes for each isomer, as shown in Figure 2.

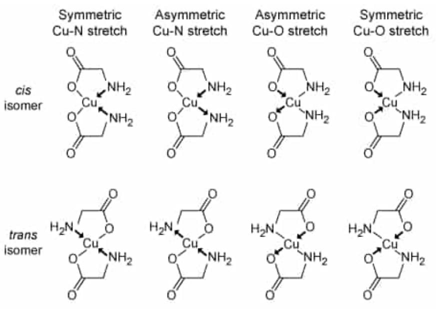


Figure 2 Vibrational modes of the coordinate bonds in copper(II) glycinate

If the stretching of the bonds during vibration causes a transient, unequal distribution of electron density between the atoms within the bond, an oscillating dipole will form. Bonds with dipoles that are not canceled out by another dipole in the molecule will absorb infrared light at particular frequencies (often expressed in wave numbers) and are designated as IR-active. As a result, the vibrational differences between the two isomers can be analyzed by far-infrared (far-IR) spectroscopy, as shown in Figure 3.

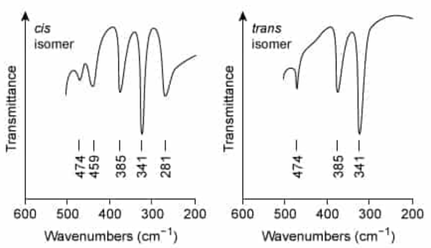


Figure 3 Far-IR spectra of the *cis* and *trans* isomers of copper(II) glycinate

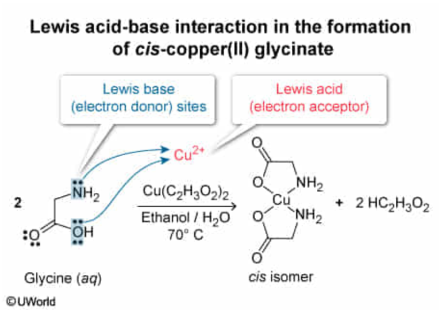
Both isomers show a skeletal absorbance (unrelated to coordinate bond vibration) near 385 cm^{-1}. Additional absorbance bands are seen that specifically relate to the vibrational modes of the bonds in each isomer.

In the reaction to form the *cis* isomer shown in Figure 1, which of the following structural sites function(s) as a Lewis base?

- A. Cu only [14%]
- B. N only [12%]
- C. O only [13%]
- ✔ D. N and O only [60%]

Correct
59% answered correctly

Explanation:



Coordination complexes consist of a central metal ion surrounded by one or more ions or molecules called **ligands**, which are bound to the metal center by coordinate bonds. The ligands act as a Lewis base (electron pair donor), and the metal center acts as a Lewis acid (electron pair acceptor). Ligand structures that can form more than one coordinate bond by donating more than one pair of electrons have more than one site that can function as a Lewis base.

During the formation of copper(II) glycinate (Figure 1), glycine displaces acetate in $Cu(C_2H_3O_2)_2$ because stronger Lewis bases can displace weaker Lewis bases as ligands within a complex. On the periodic table, Lewis base strength tends to increase moving up a column and decrease moving from left to right across a row. As such, N tends to act as a stronger Lewis base than O.

As a ligand, glycine has an available pair of electrons to donate on both the oxygen atom in the carboxylic group and on the nitrogen atom of the amino group, and both sites in the structure of glycine function as Lewis bases (electron pair donor) in the reaction shown in Figure 1. This is indicated by the formation of the Cu-N and Cu-O coordinate bonds in the complex.

(Choice A) In the reaction, copper is present in the form of Cu^{2+} ions that are supplied by the $Cu(C_2H_3O_2)_2$ solution.

The Cu^{2+} ion is electron-deficient and acts as a Lewis *acid* (electron pair *acceptor*).

(Choices B and C) The oxygen atom in the carboxylic acid group and the nitrogen atom in the amino group in the structure of glycine can both donate a lone pair of electrons. As such, both sites can function as Lewis bases.

Educational objective:
Within coordination complexes, a ligand acts as a Lewis base, and the metal center acts as a Lewis acid. Ligand structures that can form more than one coordinate bond by donating more than one pair of electrons have more than one site that can function as a Lewis base.

Time spent:0 sec
Subject:
General Chemistry

QID:402280
System:
Interactions of Chemical Substances

Amino acids and copper both serve vital roles in many biological processes of living organisms. Beyond their separate biological roles, however, chemists have also discovered that amino acids are able to function as chelates with Cu^{2+} ions to form blue complexes with potential medicinal applications. One such complex, copper(II) glycinate, has shown to be effective against some oral diseases. The reaction to form copper(II) glycinate is shown in Figure 1.

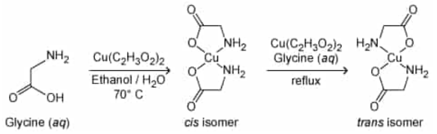


Figure 1 Synthesis of two isomers of copper(II) glycinate

Two different isomers can form depending on the reaction conditions. Mixing hot aqueous glycine (70 °C) with a hot solution of copper(II) acetate dissolved in 50% aqueous ethanol, followed by cooling the mixture on ice, yields a precipitate of only the *cis* isomer. If a suspension of the *cis* isomer is held at boiling in an aqueous solution of glycine and copper(II) acetate, the *trans* isomer is obtained.

Because of the geometric differences between the two isomers, the stretching vibrations of the central Cu–N and Cu–O bonds in each isomer are different. Symmetric and asymmetric bond stretching give four possible vibrational modes for each isomer, as shown in Figure 2.

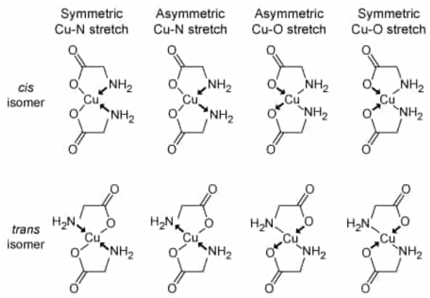


Figure 2 Vibrational modes of the coordinate bonds in copper(II) glycinate

If the stretching of the bonds during vibration causes a transient, unequal distribution of electron density between the atoms within the bond, an oscillating dipole will form. Bonds with dipoles that are not canceled out by another dipole in the molecule will absorb infrared light at particular frequencies (often expressed in wave numbers) and are designated as IR-active. As a result, the vibrational differences between the two isomers can be analyzed by far-infrared (far-IR) spectroscopy, as shown in Figure 3.

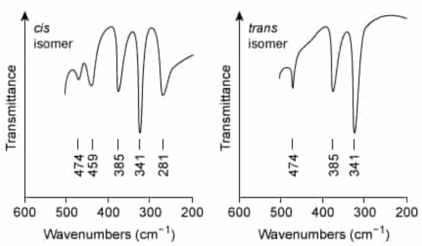


Figure 3 Far-IR spectra of the *cis* and *trans* isomers of copper(II) glycinate

Both isomers show a skeletal absorbance (unrelated to coordinate bond vibration) near 385 cm^{-1}. Additional absorbance bands are seen that specifically relate to the vibrational modes of the bonds in each isomer.

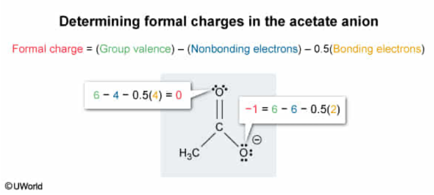
What are the formal charges of the oxygen atoms in an acetate ion isolated from the $Cu(C_2H_3O_2)_2$ used during the reaction if the effects from resonance are excluded?

- A. 0 and 0 [19%]
- B. −2 and 0 [17%]
- ✔ C. 0 and −1 [52%]
- D. −1 and −1 [10%]

Correct

52% answered correctly

Explanation:



A **formal charge** is assigned to an atom based on its bonding configuration by using an **electron accounting** method that assumes that the **bonding electrons** are shared equally between two atoms. In the Lewis structure of a molecule, any **nonbonding electrons** (lone pairs) of an atom are considered to be retained by the atom, but bonding electrons are evenly divided between two bonded atoms. The allocated electrons are then assessed relative to an atom's **group valence** (the number of valence electrons) held by an uncharged atom within a given group).

Formal charge is expressed mathematically as:

$$\begin{matrix}\text{Formal}\\\text{charge}\end{matrix} = \begin{matrix}\text{Group}\\\text{valence}\end{matrix} - \begin{matrix}\text{Nonbonding}\\\text{electrons}\end{matrix} - \frac{1}{2}\begin{pmatrix}\text{Bonding}\\\text{electrons}\end{pmatrix}$$

In the passage, the name of $Cu(C_2H_3O_2)_2$ is stated to be copper(II) acetate. The Roman numeral (II) indicates that the copper has a charge of +2. This means that each formula unit of each acetate anion must have a net charge of −1, as in $C_2H_3O_2^-$. This also indicates that acetate possesses one additional electron beyond those contributed by the valence shell of each atom. With this information, a Lewis structure obeying the octet rule can be drawn for the acetate anion. This is an individual resonance contributor that excludes effects from delocalization.

Assessing the formal charge for each oxygen atom shows that the O atom in the C=O bond has a formal charge of 0 whereas the O atom in the C–O bond has a formal charge of −1.

(Choice A) Because the acetate anion has a net formal charge of −1, at least one O atom must have a nonzero formal charge.

(Choices B and D) If both O atoms have a formal charge of −1 or one has a formal charge of −2, the total net charge for the acetate anion would exceed −1.

Educational objective:
A formal charge is assigned to an atom based on an electron accounting method that assumes the bonding electrons between two atoms are shared equally. Formal charge equals the group valence of an atom minus both the number of nonbonding electrons and half of the number of bonding electrons.

Time spent:0 sec
Subject:
General Chemistry

QID:402281
System:
Interactions of Chemical Substances

Amino acids and copper both serve vital roles in many biological processes of living organisms. Beyond their separate biological roles, however, chemists have also discovered that amino acids are able to function as chelates with Cu^{2+} ions to form blue complexes with potential medicinal applications. One such complex, copper(II) glycinate, has shown to be effective against some oral diseases. The reaction to form copper(II) glycinate is shown in Figure 1.

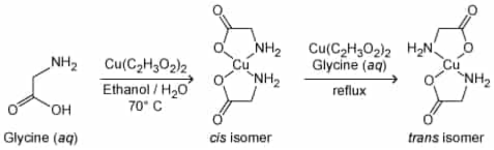


Figure 1 Synthesis of two isomers of copper(II) glycinate

Two different isomers can form depending on the reaction conditions. Mixing hot aqueous glycine (70 °C) with a hot solution of copper(II) acetate dissolved in 50% aqueous ethanol, followed by cooling the mixture on ice, yields a precipitate of only the *cis* isomer. If a suspension of the *cis* isomer is held at boiling in an aqueous solution of glycine and copper(II) acetate, the *trans* isomer is obtained.

Because of the geometric differences between the two isomers, the stretching vibrations of the central Cu–N and Cu–O bonds in each isomer are different. Symmetric and asymmetric bond stretching give four possible vibrational modes for each isomer, as shown in Figure 2.

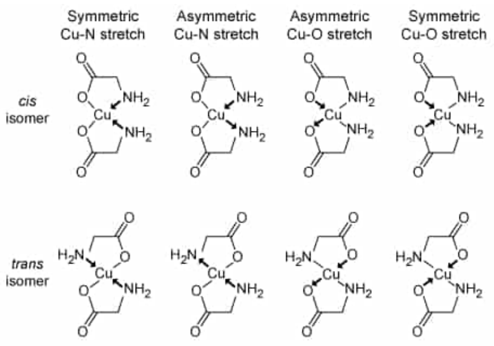


Figure 2 Vibrational modes of the coordinate bonds in copper(II) glycinate

If the stretching of the bonds during vibration causes a transient, unequal distribution of electron density between the atoms within the bond, an oscillating dipole will form. Bonds with dipoles that are not canceled out by another dipole in the molecule will absorb infrared light at particular frequencies (often expressed in wave numbers) and are designated as IR-active. As a result, the vibrational differences between the two isomers can be analyzed by far-infrared (far-IR) spectroscopy, as shown in Figure 3.

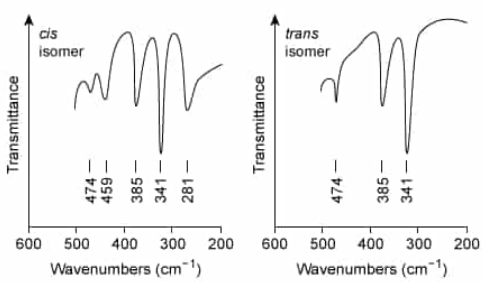


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Figure 3 Far-IR spectra of the *cis* and *trans* isomers of copper(II) glycinate

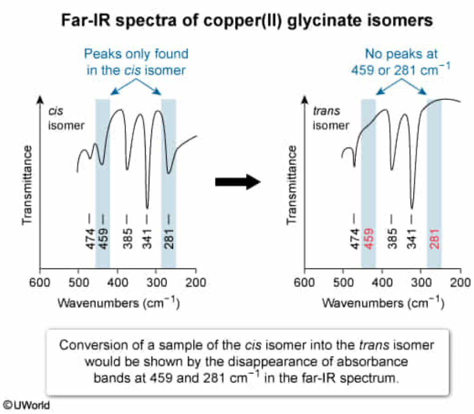
Both isomers show a skeletal absorbance (unrelated to coordinate bond vibration) near 385 cm^{-1}. Additional absorbance bands are seen that specifically relate to the vibrational modes of the bonds in each isomer.

Suppose that a small sample of the *cis* isomer precipitate was heated in the solid state in an oven at 180 °C for 30 minutes, cooled, and then analyzed by far-IR spectroscopy. If the spectrum of the thermally treated sample was similar to the spectrum of the *cis* isomer but displayed no absorbance bands at 459 and 281 cm^{-1}, the sample most likely was:

- A. a mixture of the *cis* and *trans* isomers. [8%]
- B. unchanged by heating. [3%]
- C. thermally decomposed. [10%]
- D. converted into the *trans* isomer. [77%]

Correct
76% answered correctly

Explanation:



Infrared (IR) spectroscopy can be used to **detect structural changes** within molecules during **chemical conversions** by monitoring sample spectra for the **emergence or disappearance of peaks** that are unique to a given structural feature of a molecule. For example, if a reactant has a structural feature that produces a distinct IR absorbance peak that the product of a reaction does not have, then the formation of the product is indicated in successive IR spectra taken during the reaction as the disappearance of the peak associated with the reactant.

Although IR spectroscopy is typically covered in organic chemistry, the question presented here is solely one of data analysis and does not require the types of analysis applied for organic chemistry. The far-IR spectrum of the treated sample discussed in this question is not shown, but its key features are described and can be compared to the spectra of each isomer given in the passage. Comparing the far-IR spectra of the isomers in the passage shows that the peaks at 459 and 281 cm^{-1} are unique to the *cis* isomer and are absent for the *trans* isomer.

Because the far-IR spectrum of the thermally treated sample is said to be similar to the spectrum of the *cis* isomer but lacks peaks at 459 and 281 cm^{-1}, it is most likely that the *cis* sample was converted to the *trans* isomer upon heating. This conclusion is further supported by the observation from the passage that the *cis* isomer converts to the *trans* isomer when held at an elevated temperature in solution. Therefore, a similar (faster) conversion is plausible for a sample in solid state held at an even higher temperature.

(Choice A) If some of the *cis* isomer were still present in the sample, absorbance peaks at 459 and 281 cm^{-1} would still be seen in the far-IR spectrum of the sample.

(Choice B) The disappearance of the absorbance peaks at 459 and 281 cm^{-1} indicates that the sample *has changed* in some way due to heating the sample.

(Choice C) If the thermally treated sample had decomposed, its spectrum would be unlikely to resemble the spectra of the isomers as reported.

Educational objective:
Infrared spectroscopy can be used to detect structural changes within molecules during chemical conversions by monitoring successive sample spectra for the emergence or disappearance of peaks that are unique to a given structural feature of a molecule.

Time spent:0 sec
Subject:
General Chemistry

QID:402282
System:
Interactions of Chemical Substances

Amino acids and copper both serve vital roles in many biological processes of living organisms. Beyond their separate biological roles, however, chemists have also discovered that amino acids are able to function as chelates with Cu^{2+} ions to form blue complexes with potential medicinal applications. One such complex, copper(II) glycinate, has shown to be effective against some oral diseases. The reaction to form copper(II) glycinate is shown in Figure 1.

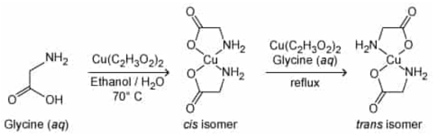


Figure 1 Synthesis of two isomers of copper(II) glycinate

Two different isomers can form depending on the reaction conditions. Mixing hot aqueous glycine (70 °C) with a hot solution of copper(II) acetate dissolved in 50% aqueous ethanol, followed by cooling the mixture on ice, yields a precipitate of only the *cis* isomer. If a suspension of the *cis* isomer is held at boiling in an aqueous solution of glycine and copper(II) acetate, the *trans* isomer is obtained.

Because of the geometric differences between the two isomers, the stretching vibrations of the central Cu–N and Cu–O bonds in each isomer are different. Symmetric and asymmetric bond stretching give four possible vibrational modes for each isomer, as shown in Figure 2.

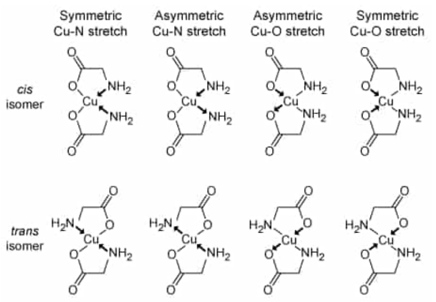


Figure 2 Vibrational modes of the coordinate bonds in copper(II) glycinate

If the stretching of the bonds during vibration causes a transient, unequal distribution of electron density between the atoms within the bond, an oscillating dipole will form. Bonds with dipoles that are not canceled out by another dipole in the molecule will absorb infrared light at particular frequencies (often expressed in wave numbers) and are designated as IR-active. As a result, the vibrational differences between the two isomers can be analyzed by far-infrared (far-IR) spectroscopy, as shown in Figure 3.

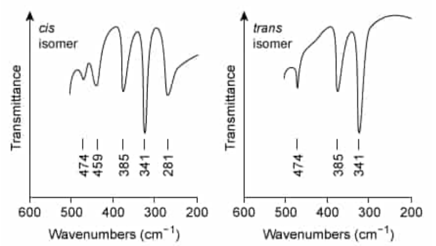


Figure 3 Far-IR spectra of the *cis* and *trans* isomers of copper(II) glycinate

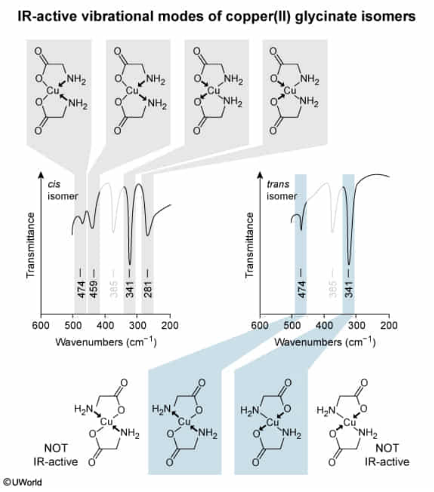
Both isomers show a skeletal absorbance (unrelated to coordinate bond vibration) near 385 cm^{-1}. Additional absorbance bands are seen that specifically relate to the vibrational modes of the bonds in each isomer.

How many vibrational modes of the coordinate bonds from each isomer of copper(II) glycinate are observable by far-IR spectroscopic analysis? (Note: Exclude the peak at 385 cm^{-1} caused by skeletal vibrational modes of the C–C and C–H bonds that are present in both isomers and are unrelated to the coordinate bonds.)

- A. Four modes in both the *trans* isomer and the *cis* isomer [6%]
- ✔ B. Two modes in the *trans* isomer and 4 modes in the *cis* isomer [80%]
- C. Four modes in the *trans* isomer and 2 modes in the *cis* isomer [7%]
- D. Three modes in the *trans* isomer and 5 modes in the *cis* isomer [6%]

Correct
80% answered correctly

Explanation:



Atoms within a molecule exhibit continual vibrational motion that can occur in a variety of bond stretching and bending modes. If the vibrational motion of a bond causes a transient, unequal distribution of electron density between the atoms within the bond, an oscillating dipole will form. If a **net change** in a **dipole** is produced by a particular **vibrational mode** and is not cancelled out by another dipole in the molecule, the vibration is **IR-active** and will absorb IR light at particular frequencies proportional to the frequency of the bond vibration. Far-IR spectroscopy measures absorbances below 600 cm^{-1}.

Because of the geometric differences between the two isomers of copper(II) glycinate, the stretching vibrations of the central Cu–N and Cu–O bonds in each isomer are different. Symmetric and asymmetric bond stretching give four possible vibrational modes for these bonds in each isomer. If all four modes were IR-active in each isomer, the **far-IR spectrum** of both isomers would show four **absorbance peaks** attributed to these vibrational modes.
Comparing the far-IR spectra in Figure 3, the *cis* isomer displays four peaks (in addition to the skeletal vibration peak at 385 cm^{-1}); however, the *trans* isomer displays only two peaks along with the skeletal vibration peak. This indicates that two of the four vibrational modes of the coordinate bonds in the *trans* isomer are not IR-active (ie, the dipoles are oriented in opposite directions that cancel and yield no net dipole).

Therefore, there are four vibrational modes from the coordinate bonds in the *cis* isomer and two vibrational modes from the coordinate bonds in the *trans* isomer that are observable by far-IR spectroscopy.

(Choices A and C) In the far-IR spectra of the isomers, the *trans* isomer displays two peaks (not four), and the *cis* isomer displays four peaks (not two) in addition to the skeletal absorbance peak.

(Choice D) In the far-IR spectra of the isomers, the *trans* isomer displays a total of three peaks and the *cis* isomer displays a total of five peaks. (The skeletal absorbance peak at 385 cm^{-1} is unrelated to the coordinate bond stretching modes and must be excluded.)

Educational objective:
Vibrational modes that do not produce a net change in a dipole are not IR-active and will not produce an absorbance peak in an IR spectrum.

Time spent:0 sec
Subject:
General Chemistry

QID:402283
System:
Interactions of Chemical Substances